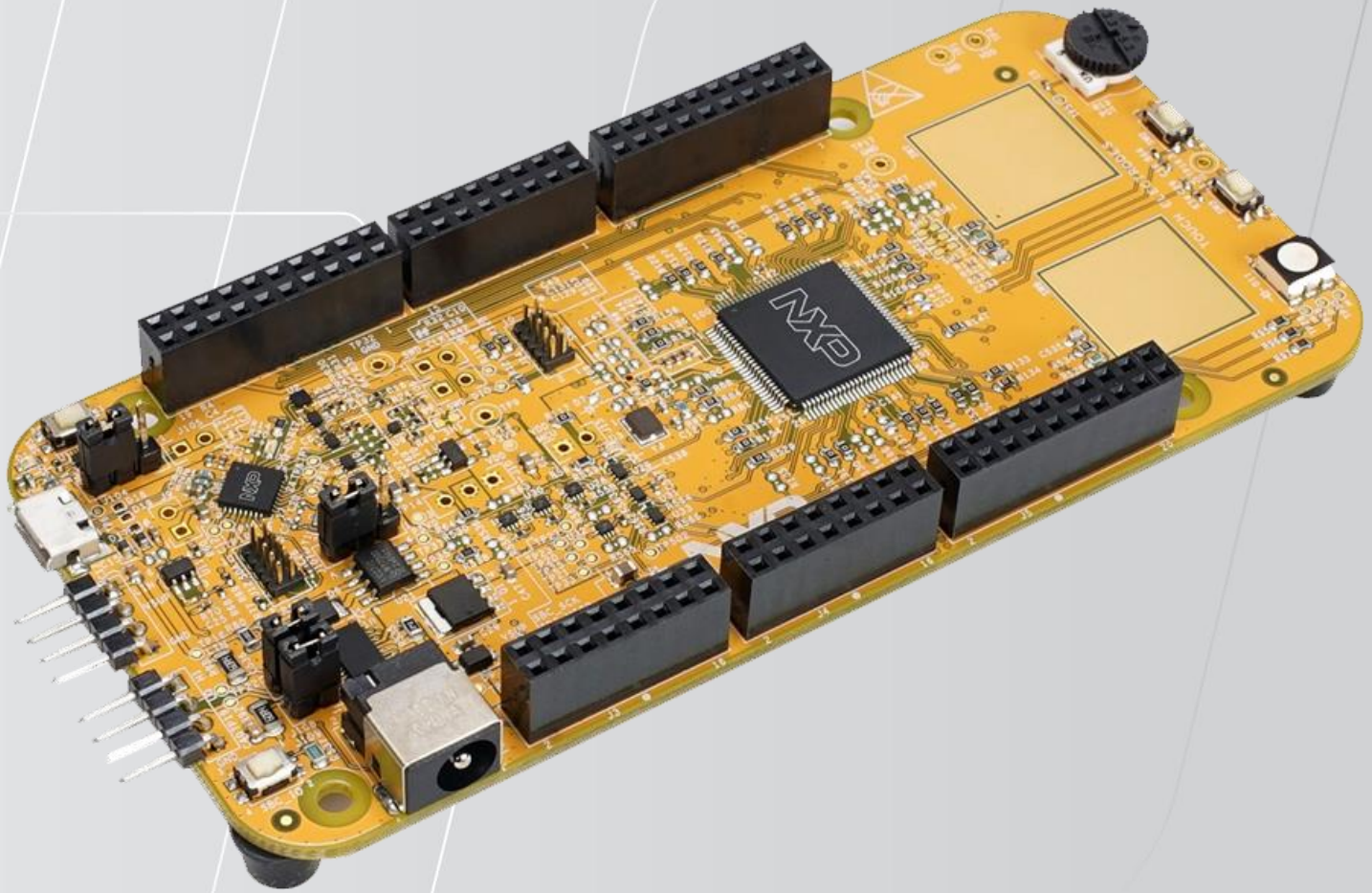
UART

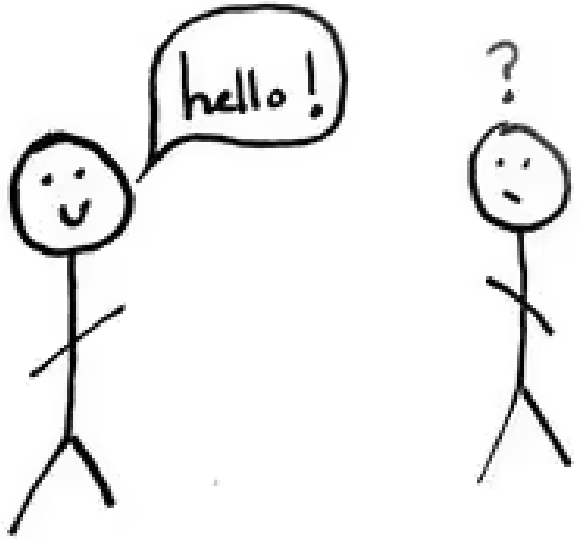


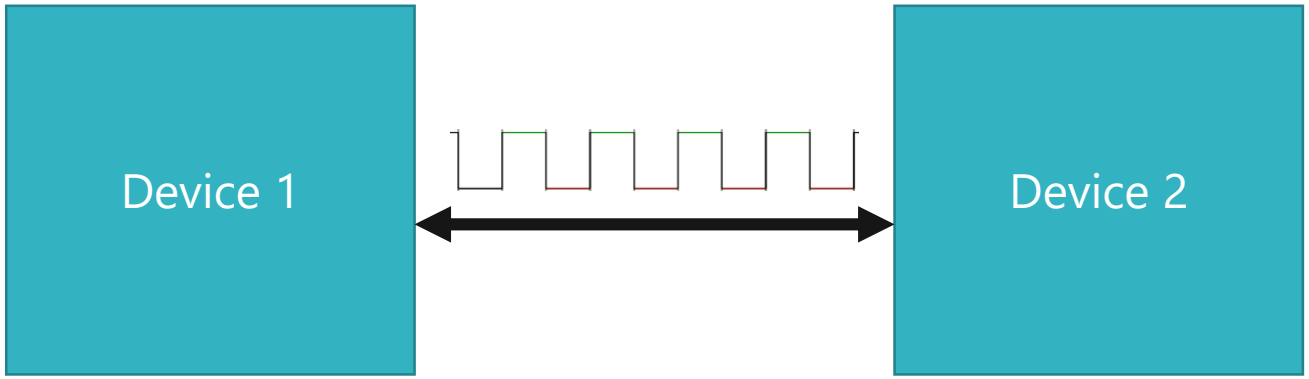
1 Dec 2022

OUTLINE

1. Communication overview
2. UART overview
3. LPUART
4. LPUART Interrupt

1. Communication overview

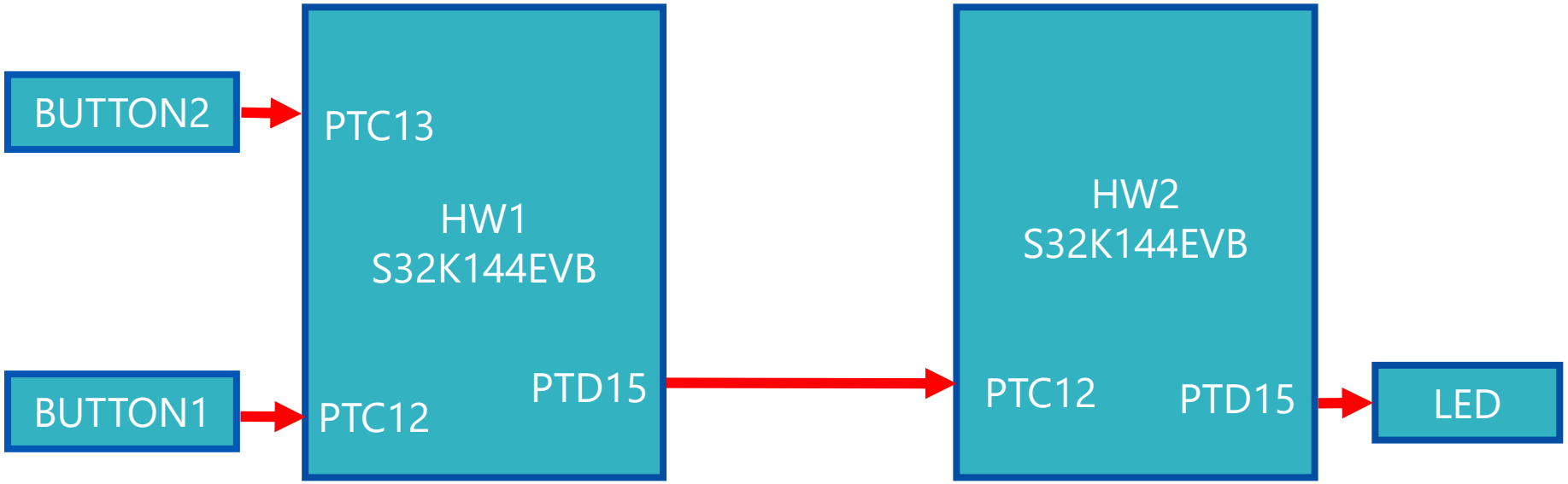




1. Communication overview

Requirement: Using buttons at HW1 to change led state at HW2 via GPIO

- if button 1 at HW1 is pressed then led at HW2 is on
- if button 2 at HW1 is pressed then led at HW2 is off



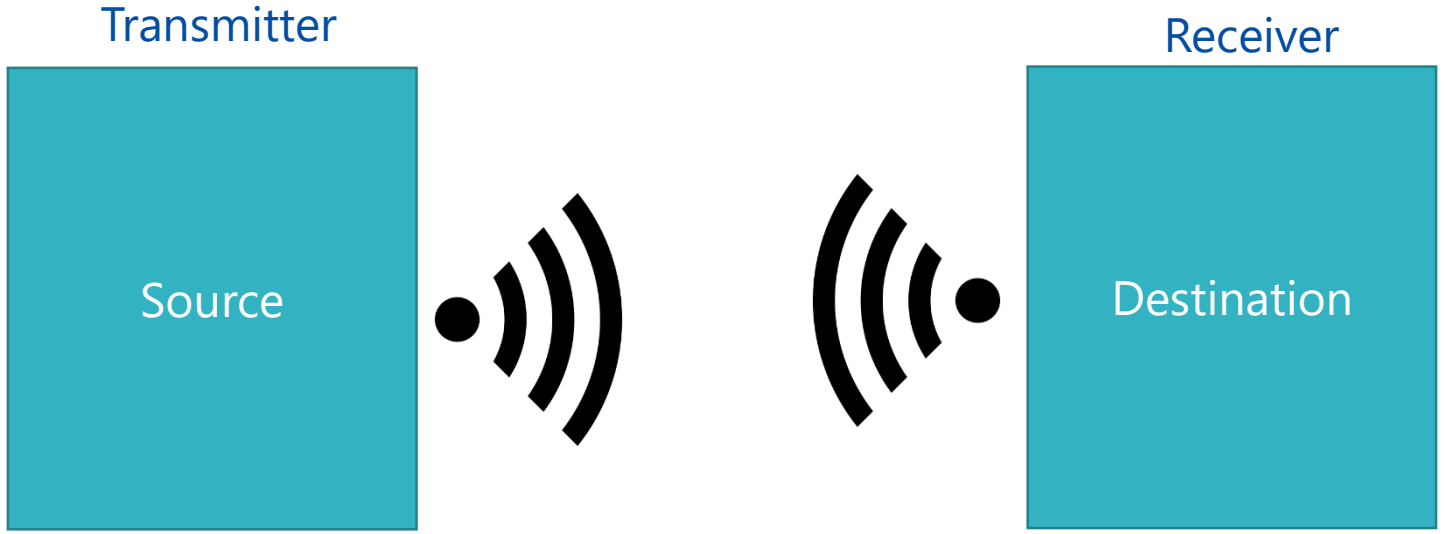
Connection Diagram

1. Communication overview

Wired

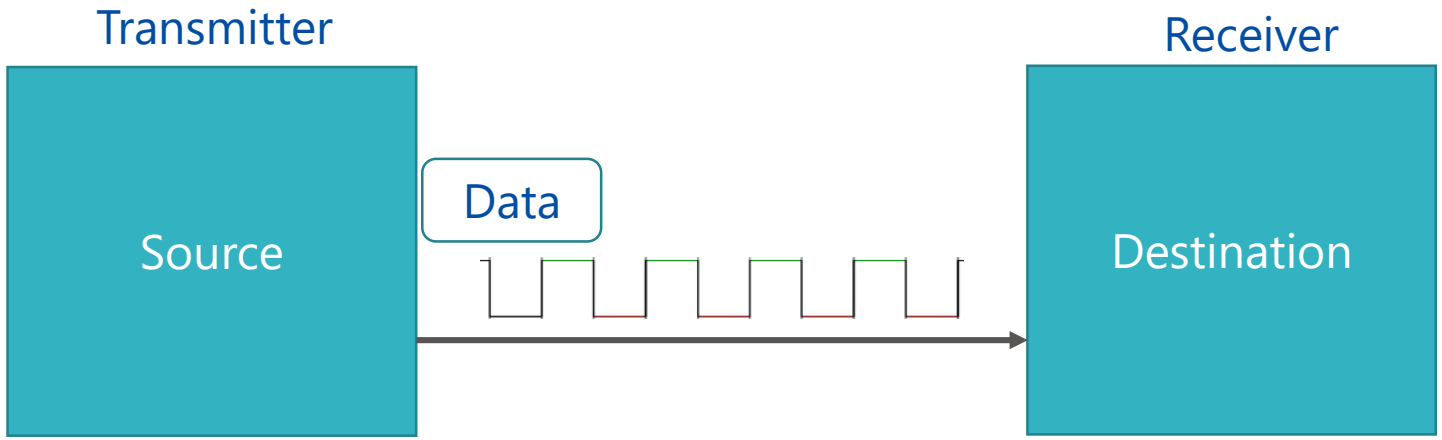


Wireless

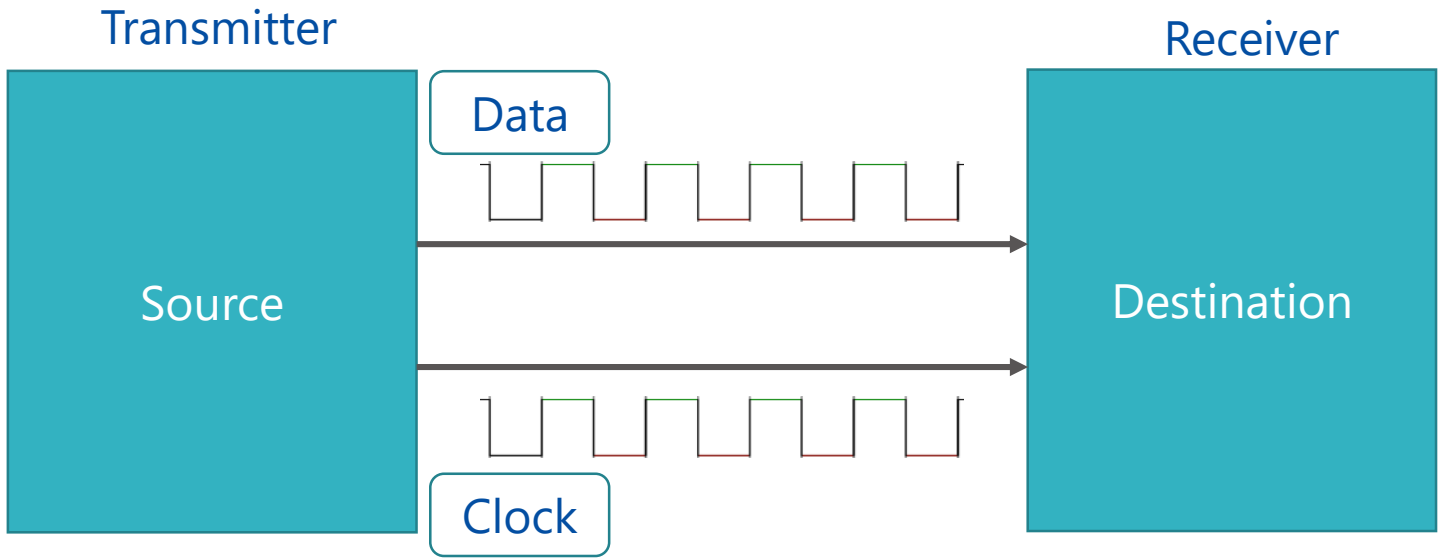


1. Communication overview

Asynchronous

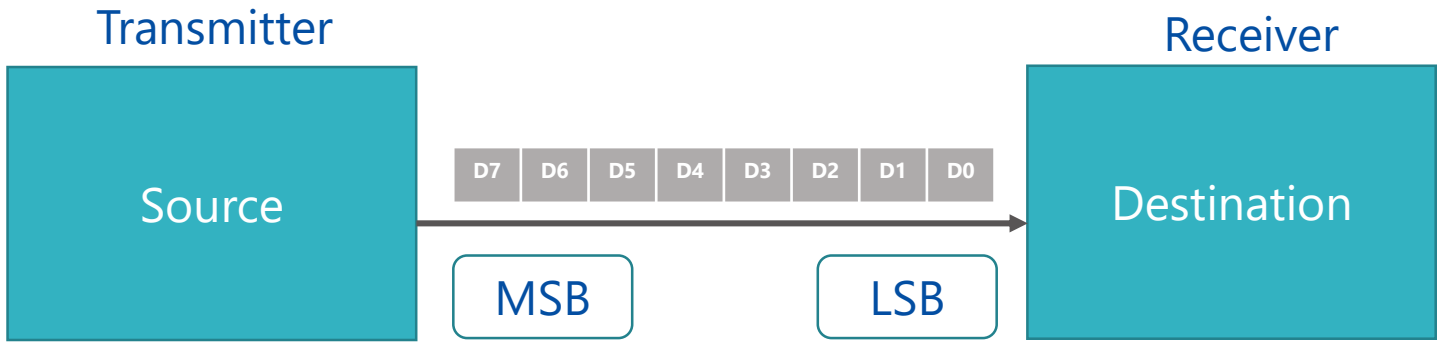


Synchronous

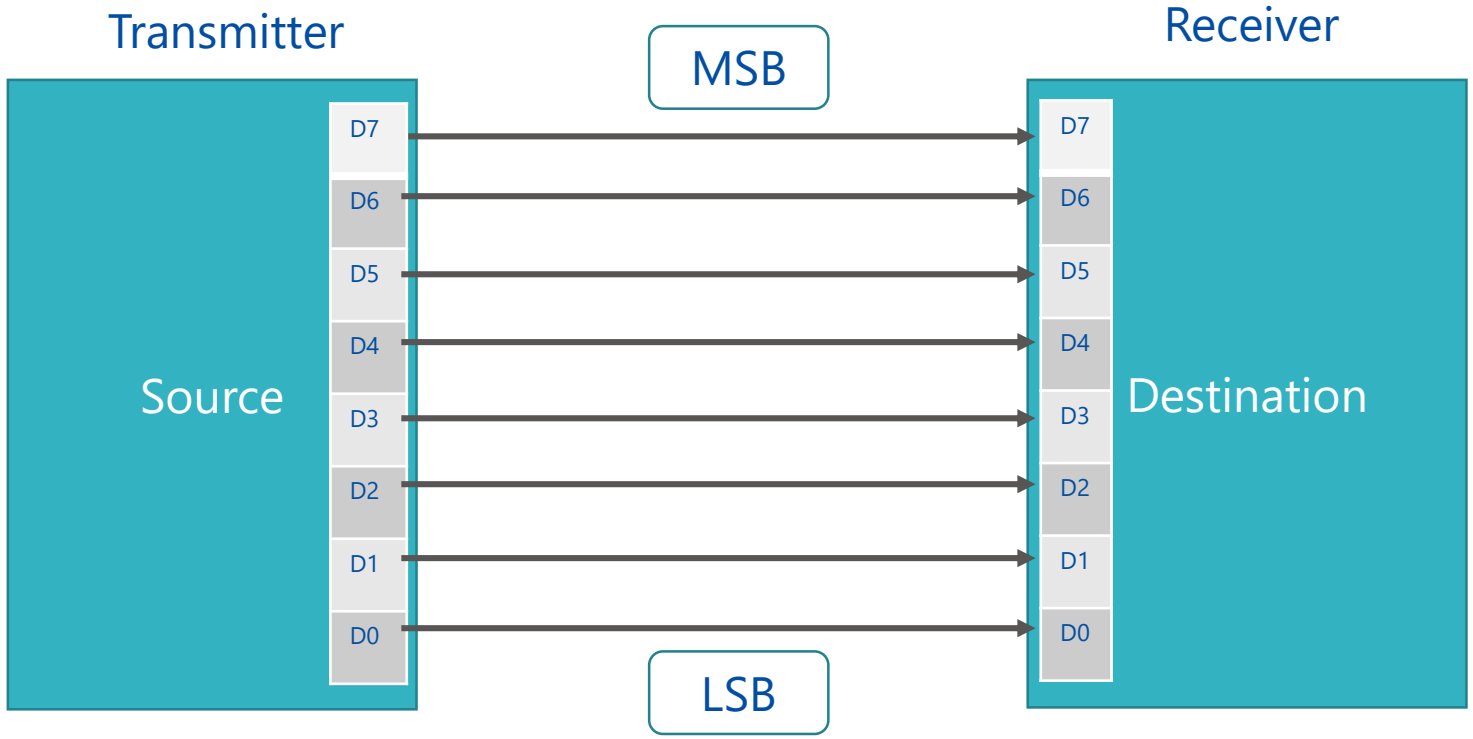


1. Communication overview

Serial

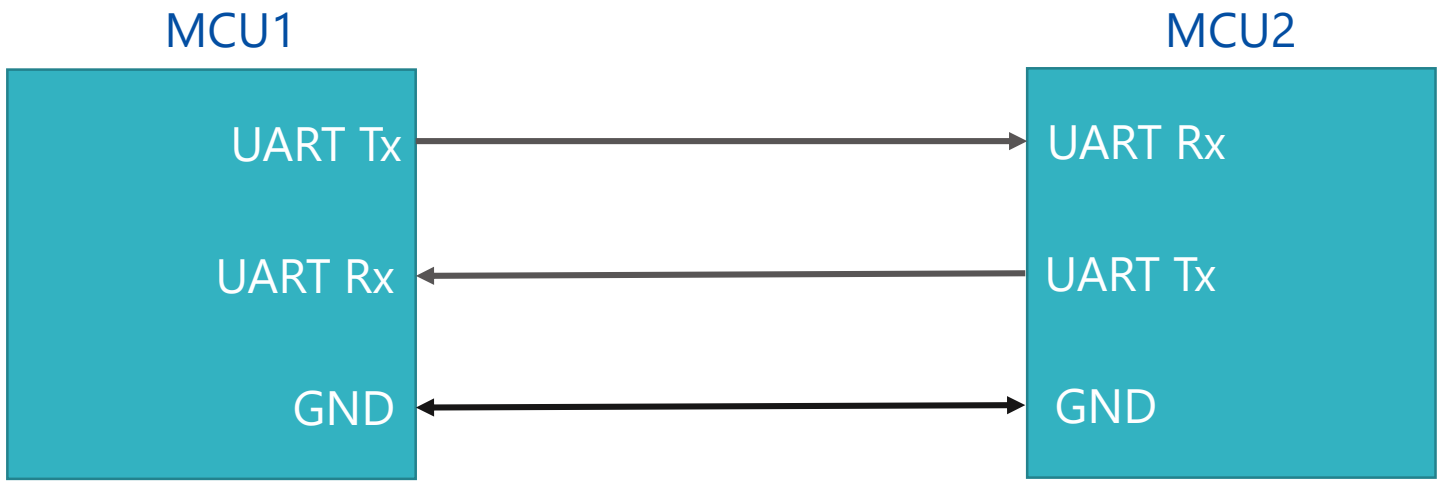


Parallel



2. UART overview

- Universal Asynchronous Receiver/ Transmitter
- Protocol for exchanging serial data between two devices
- Uses only two wired
  - Tx to Rx (each direction)
- Can be simplex, half-duplex or full-duplex
- Data is transmitted as frames



2. UART overview

Where is UART used?

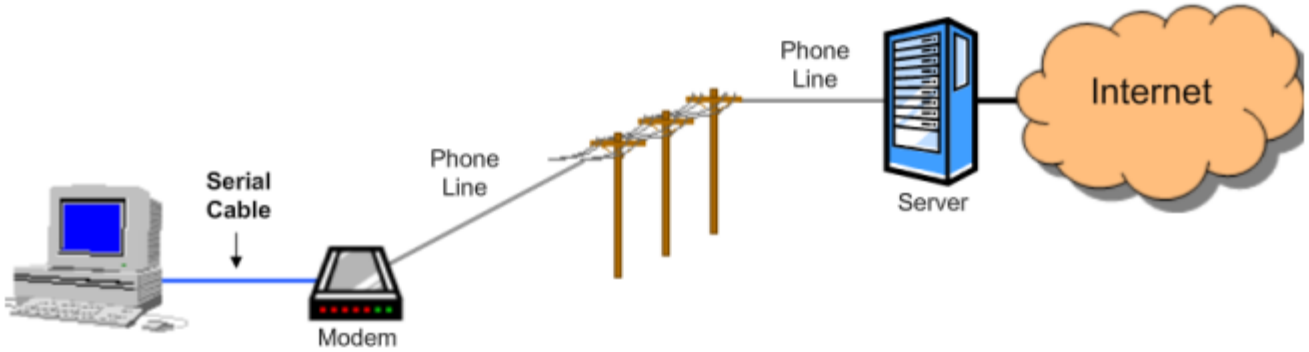
- Serial(COM) port
- RS-232
- RS-485



PC to Device



PC to PC

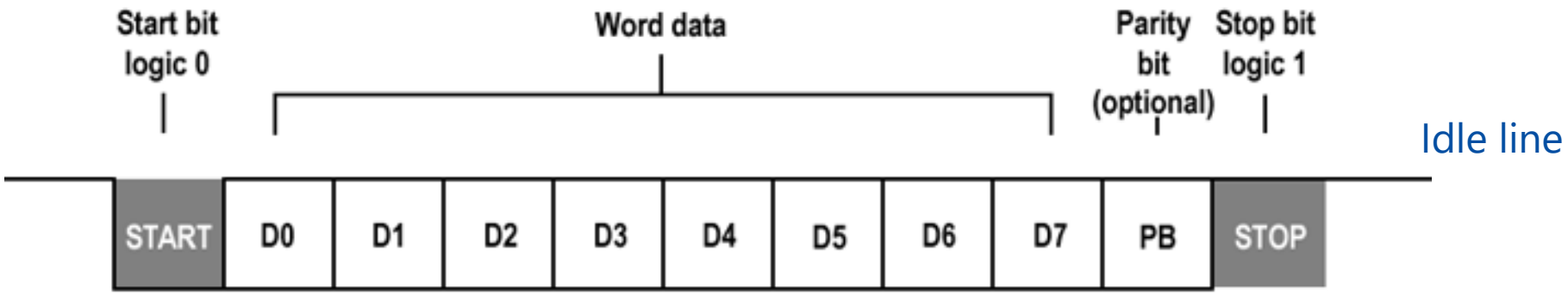


PC to Internet



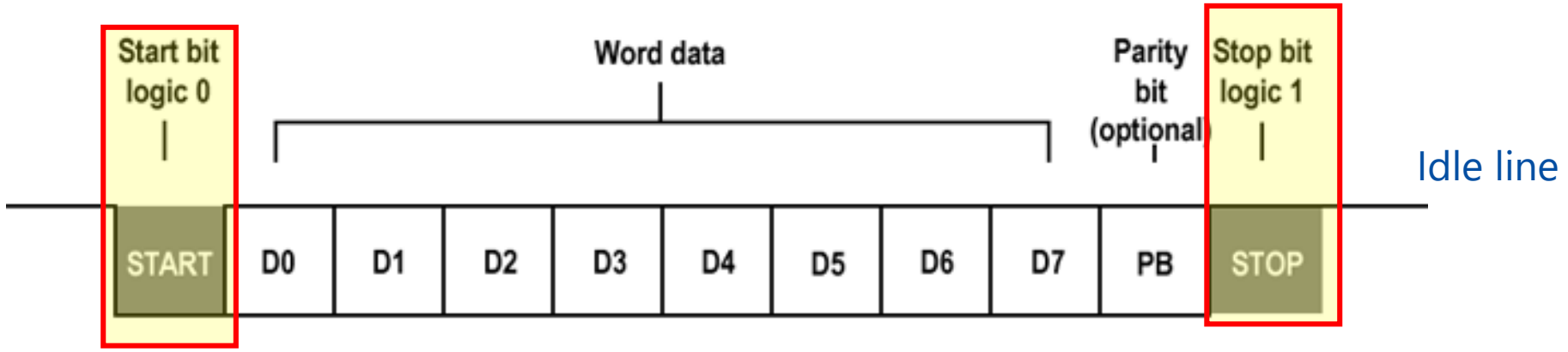
2.1. UART frame format

- UART frames consist of:
  - Start/Stop bits
  - Data bits
  - Parity bit(optional)
- High voltage(mark=1), low voltage(space=0)
- In the idle state, the line is held high



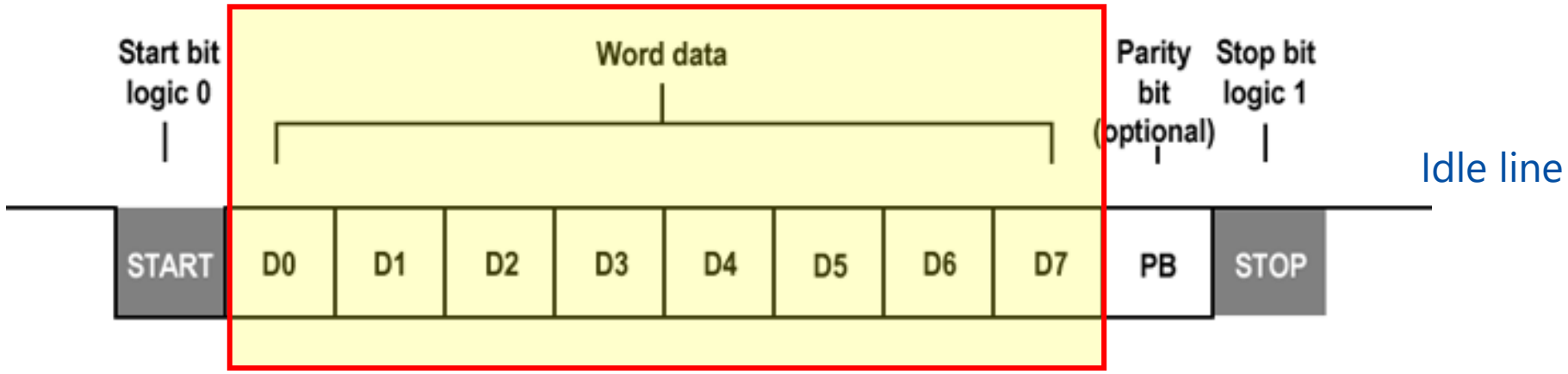
2.2. Start and stop bits

- The start bit indicates data is coming
  - Transition from idle (high) to low
- The stop bit(s) indicate data is complete
  - Stay/ Return to idle(high)
  - Second(optional) stop bit



2.3. Data bits

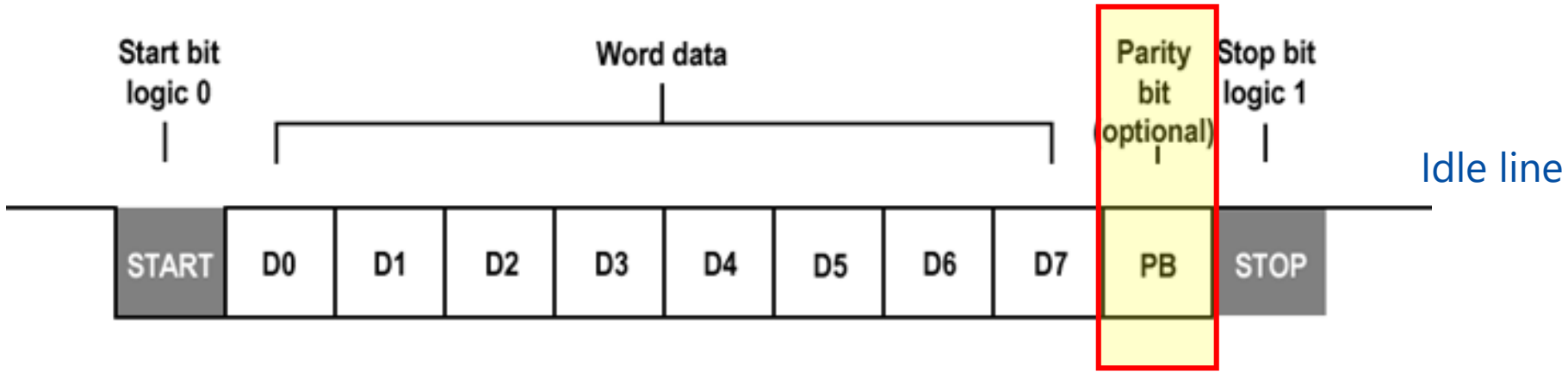
- User data
- Length: 5 to 9 (usually 7 or 8)
- Data is typically sent with the least significant bit (LSB)



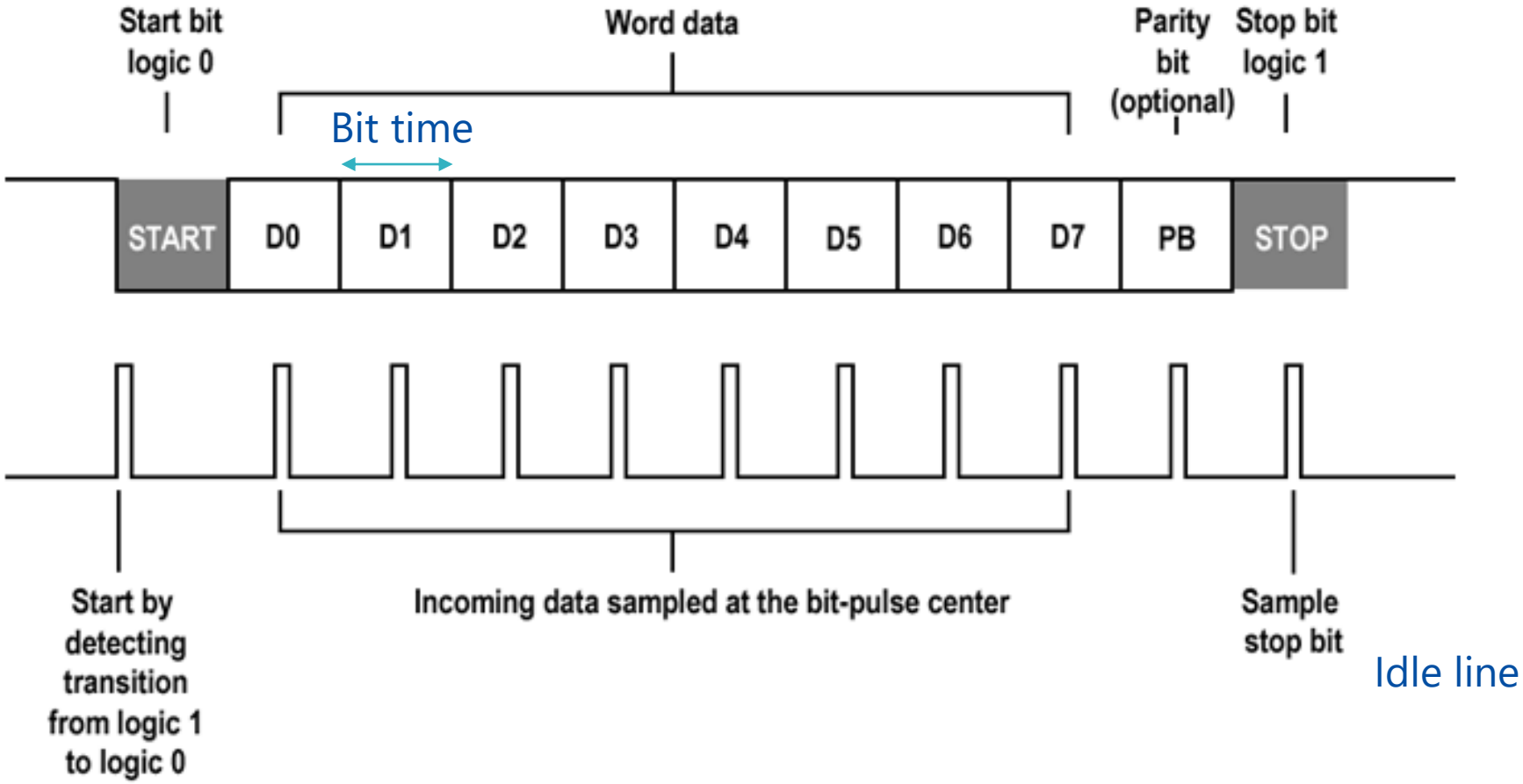
2.4. Parity bit(optional)

- Used for error detection
- Even parity: number of 1's must be even
- Odd parity: number of 1's must be odd

Example: ASCII 'H'(0x48= 1 0 0 1 0 0 0) = two 1's



2.5. UART Timing



The baud rate is the rate at which information is transferred in a communication channel. In the serial port context, "9600 baud" means that the serial port is capable of transferring a maximum of 9600 bits per second.

2.6. UART Data throughput

Data throughput Example

+ Assume 19200 baud rate , 8 data bits, no parity, 1 stop bit

  - Baud rate 19200 → 19.2kbps
  - 1 start bit + 8 data bits + 1 stop bit = 10 bits
  - It takes 10 bits to send 8 bits of data.
  - 19.2 kbps * 8/10 = 15.36 kbps.
  - How many KB (kilobytes) per second:

  =15.36 * 1KB / 8192 bits = 1.875 KB/s

3. Low Power Universal Asynchronous Receiver/ Transmitter (LPUART)

Features of the LPUART module include:

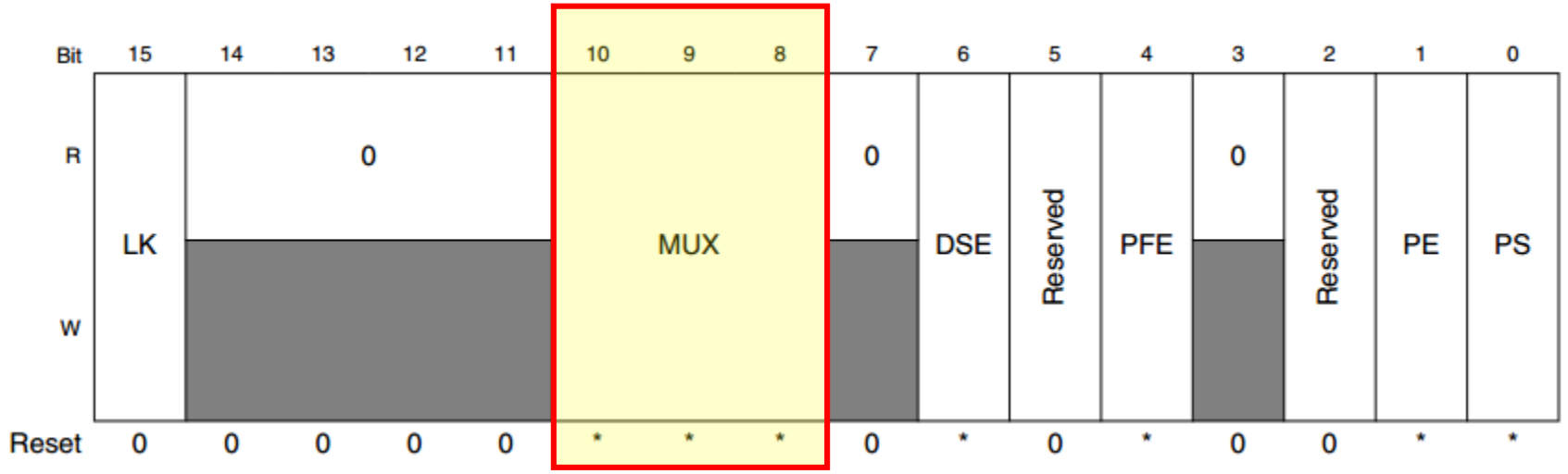
- Full-duplex, standard non-return-to-zero (NRZ) format
- Programmable baud rates (13-bit modulo divider)
- Interrupt, DMA or polled operation
- Hardware parity generation and checking
- Programmable 7-bit, 8-bit, 9-bit or 10-bit character length
- Programmable 1-bit or 2-bit stop bits
- Three receiver wakeup methods
- Configurable idle length detection
- Independent FIFO structure for transmit and receive

3.1. UART signals

53.2.3 Signal Descriptions

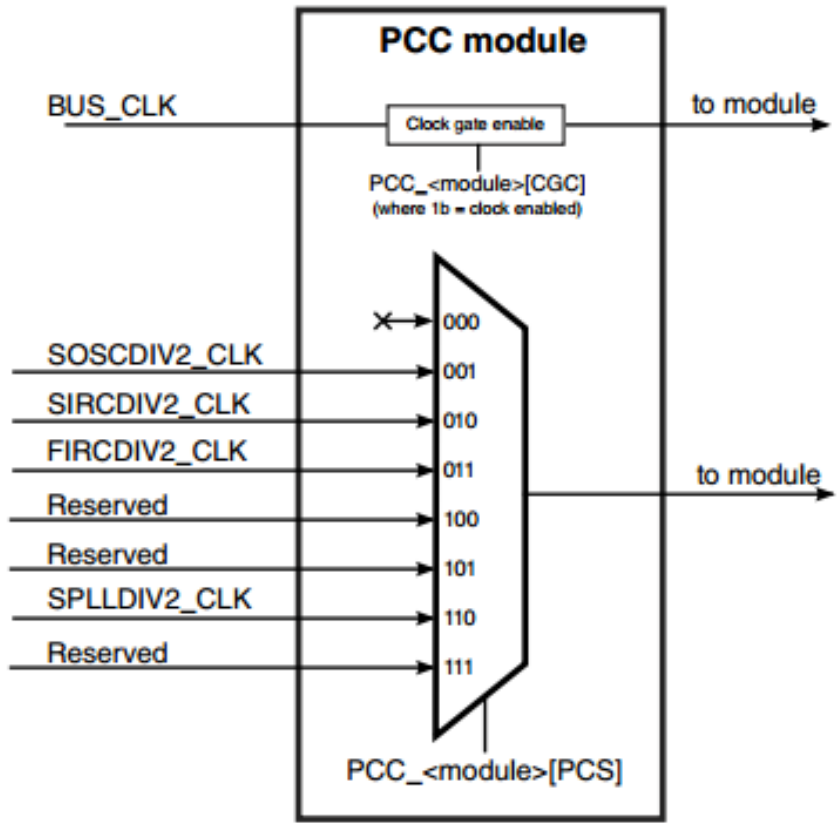
Signal	Description	I/O
TXD	Transmit data. This pin is normally an output, but is an input (tristated) in single wire mode whenever the transmitter is disabled or transmit direction is configured for receive data.	I/O
RXD	Receive data.	I
CTS_B	Clear to send.	I
RTS_B	Request to send.	O

3.2. Setting Tx/Rx pin



10–8 MUX	Pin Mux Control Not all pins support all pin muxing slots. Unimplemented pin muxing slots are reserved and may result in configuring the pin for a different pin muxing slot. The corresponding pin is configured in the following pin muxing slot as follows: 000 Pin disabled (Alternative 0) (analog). 001 Alternative 1 (GPIO). 010 Alternative 2 (chip-specific). 011 Alternative 3 (chip-specific). 100 Alternative 4 (chip-specific). 101 Alternative 5 (chip-specific). 110 Alternative 6 (chip-specific). 111 Alternative 7 (chip-specific).

3.3. Source clock LPUART

Modules	PCC multiplexer
LPUART	

3.4. Baud rate generation

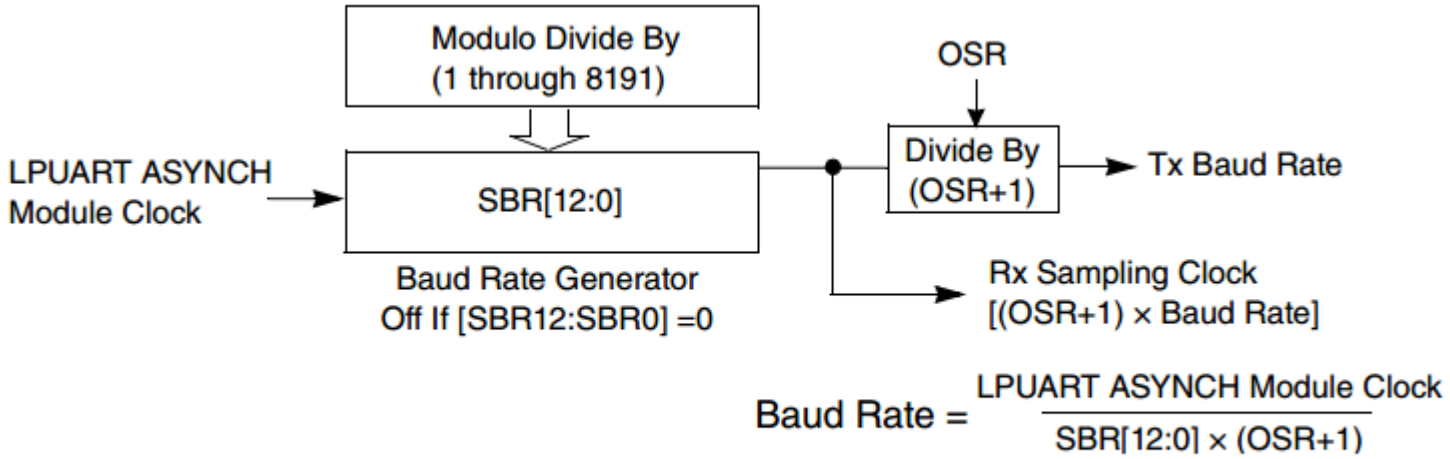


Figure 53-3. LPUART baud rate generation

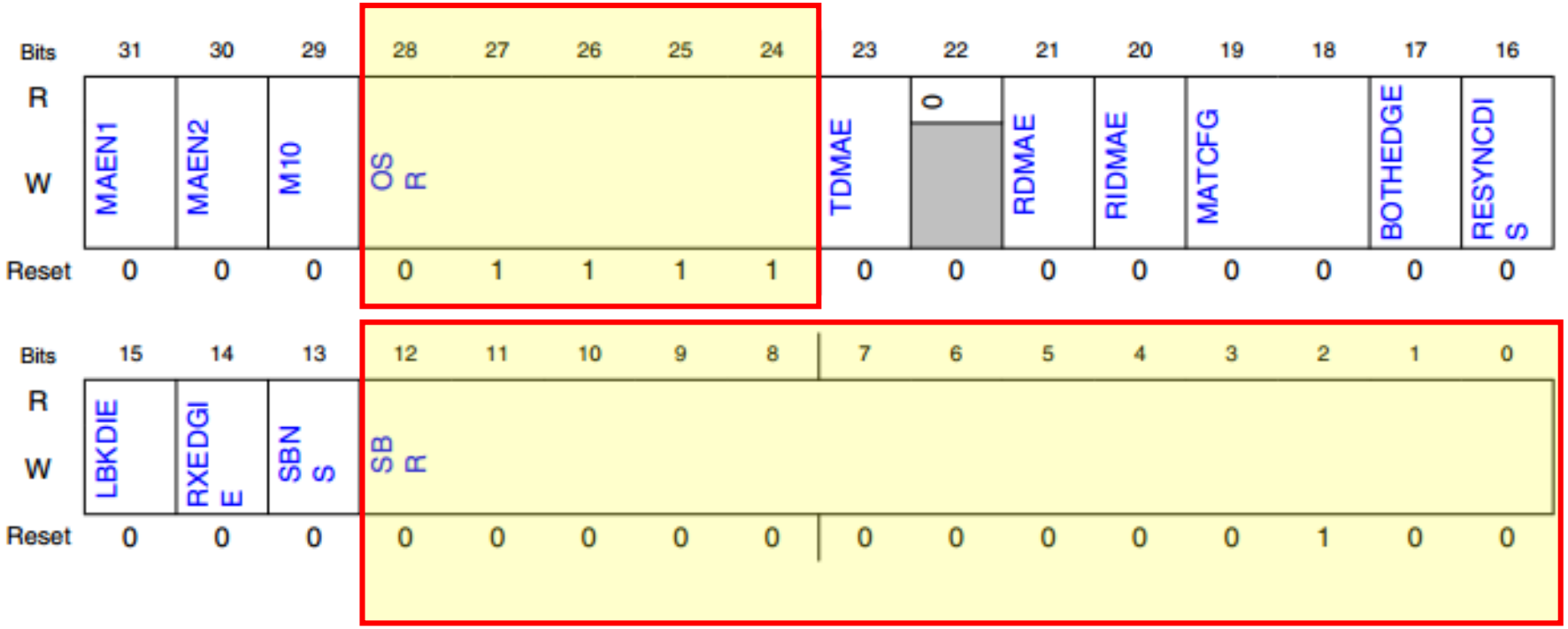
1. Oversampling Ratio: 8 → 32
2. Select baud rate: 4800, 9600, 14400, 19200, 38400, 56000, 57600, 115000
3. Calculate Baud Rate Modulo Divisor

$$\text{Baud Rate Modulo Divisor} = \frac{f_{LPUART}}{Baud\ rate * Oversampling\ Ratio}$$

3.4. Baud rate generation

LPUART Baud Rate Register (BAUD)

Oversampling Ratio



Baud Rate Modulo Divisor

3.5. Block diagram

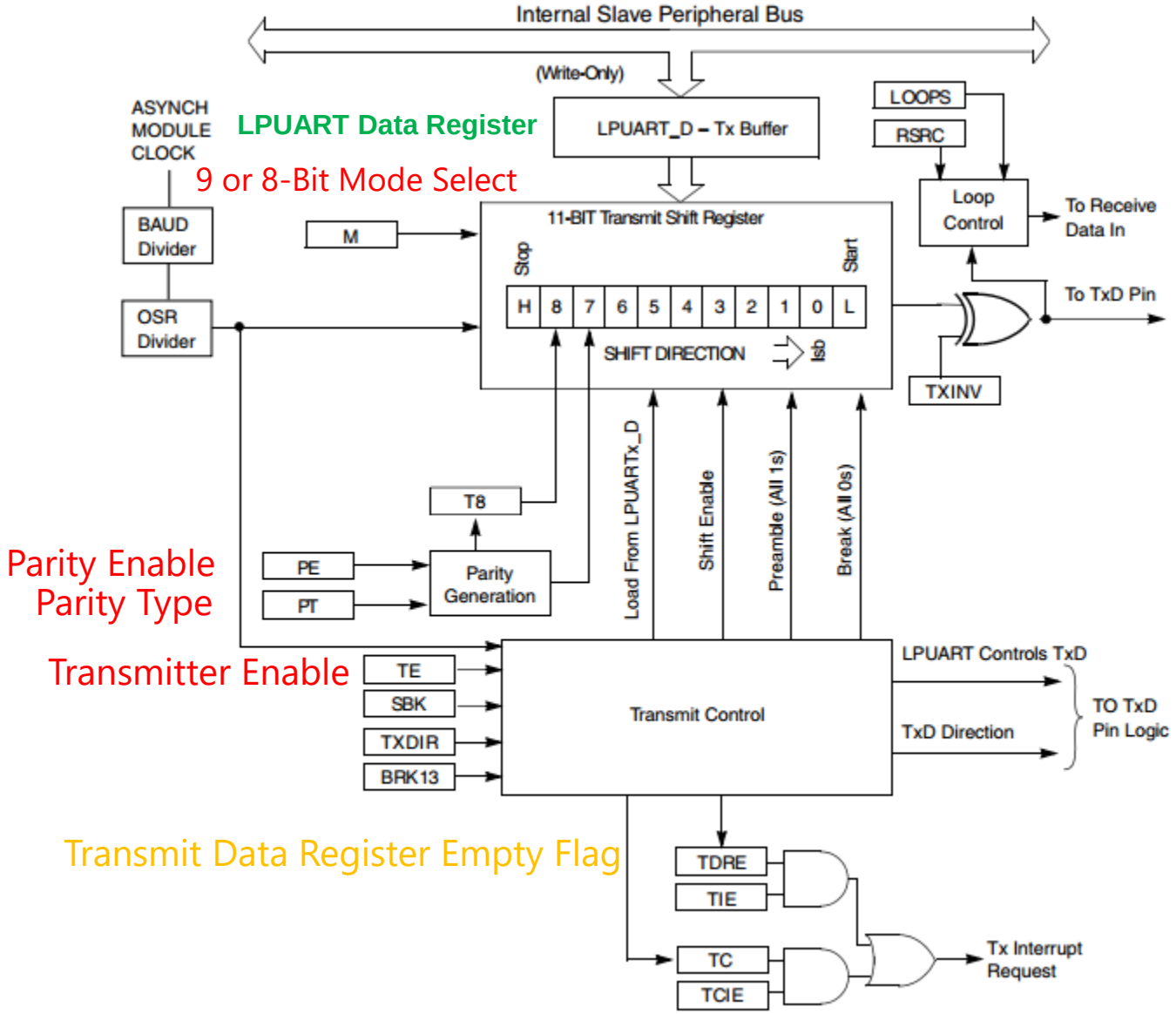


Figure 53-1. LPUART transmitter block diagram

3.5. Block diagram

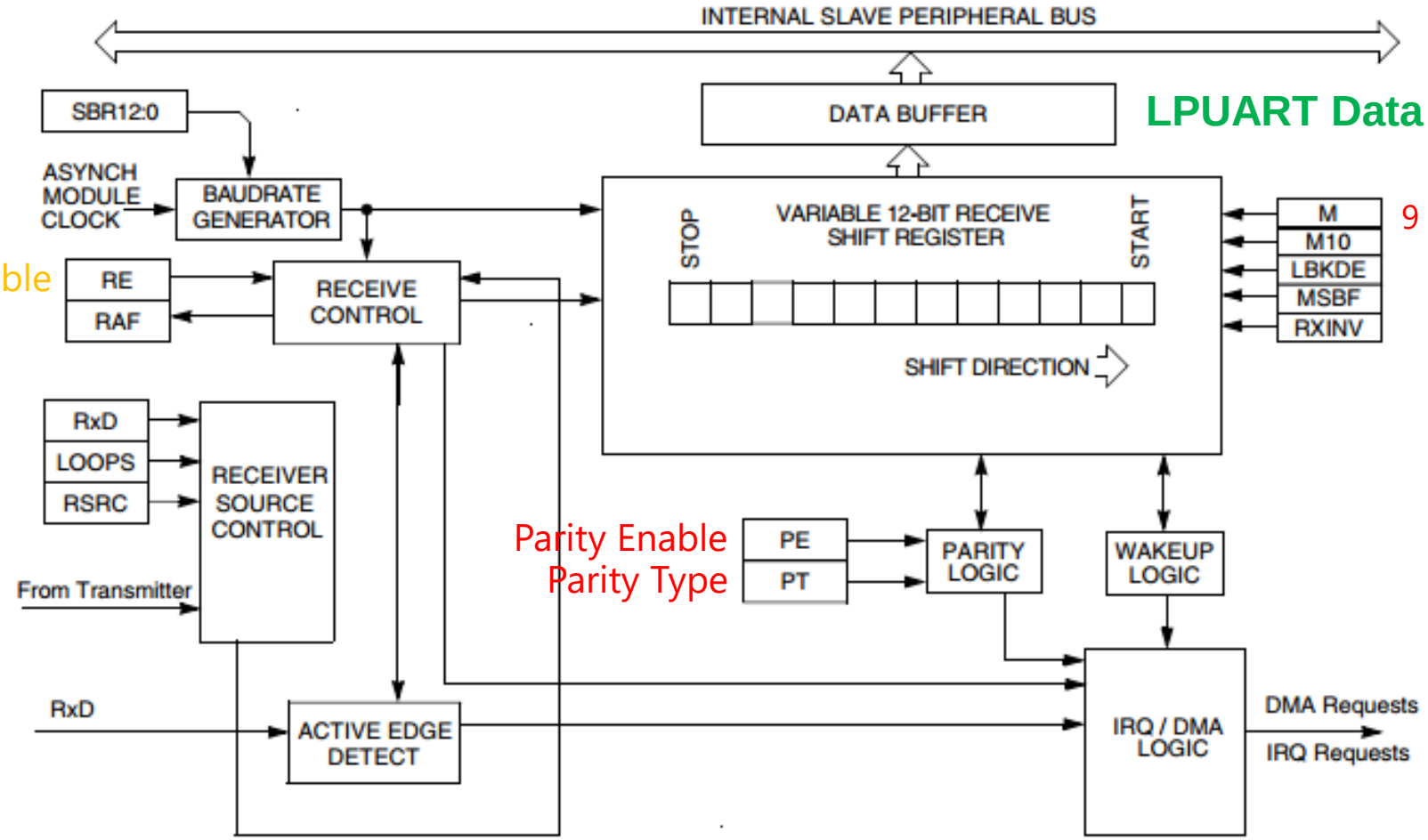


Figure 53-2. LPUART receiver block diagram

3.6. LPUART memory map

53.3.1.1 LPUART memory map

LPUART0 base address: 4006_A000h

LPUART1 base address: 4006_B000h

LPUART2 base address: 4006_C000h

Offset	Register	Width (In bits)	Access	Reset value
0h	Version ID Register (VERID)	32	RO	0401_0003h
4h	Parameter Register (PARAM)	32	RO	0000_0202h
8h	LPUART Global Register (GLOBAL)	32	RW	0000_0000h
Ch	LPUART Pin Configuration Register (PINCFG)	32	RW	0000_0000h
10h	LPUART Baud Rate Register (BAUD)	32	RW	0F00_0004h
14h	LPUART Status Register (STAT)	32	RW	00C0_0000h
18h	LPUART Control Register (CTRL)	32	RW	0000_0000h
1Ch	LPUART Data Register (DATA)	32	RW	0000_1000h
20h	LPUART Match Address Register (MATCH)	32	RW	0000_0000h
24h	LPUART Modem IrDA Register (MODIR)	32	RW	0000_0000h
28h	LPUART FIFO Register (FIFO)	32	RW	00C0_0011h
2Ch	LPUART Watermark Register (WATER)	32	RW	0000_0000h

3.6.1. Assignment 1

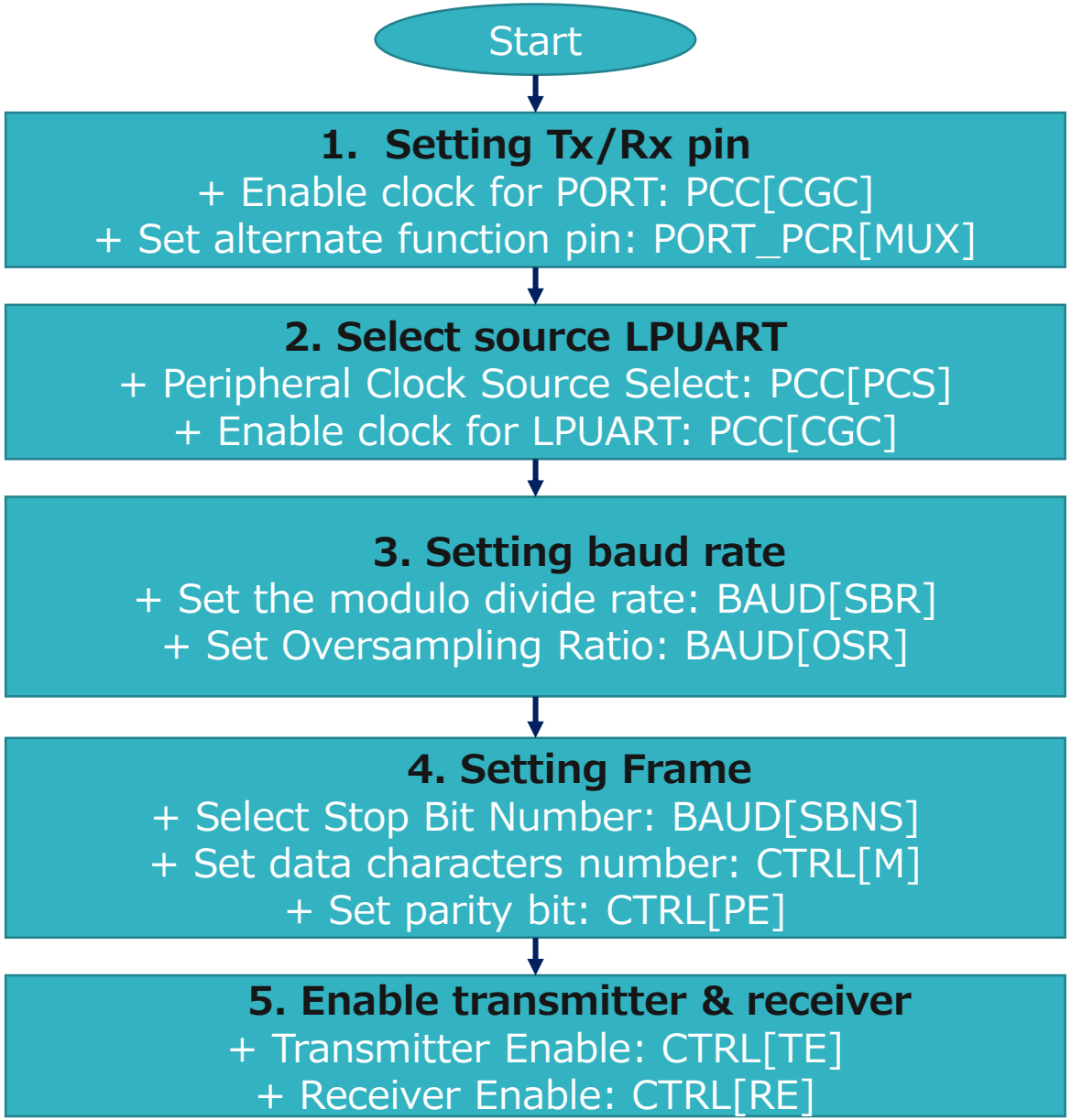
Write a define struct to manage LPUART memory map

Ex:

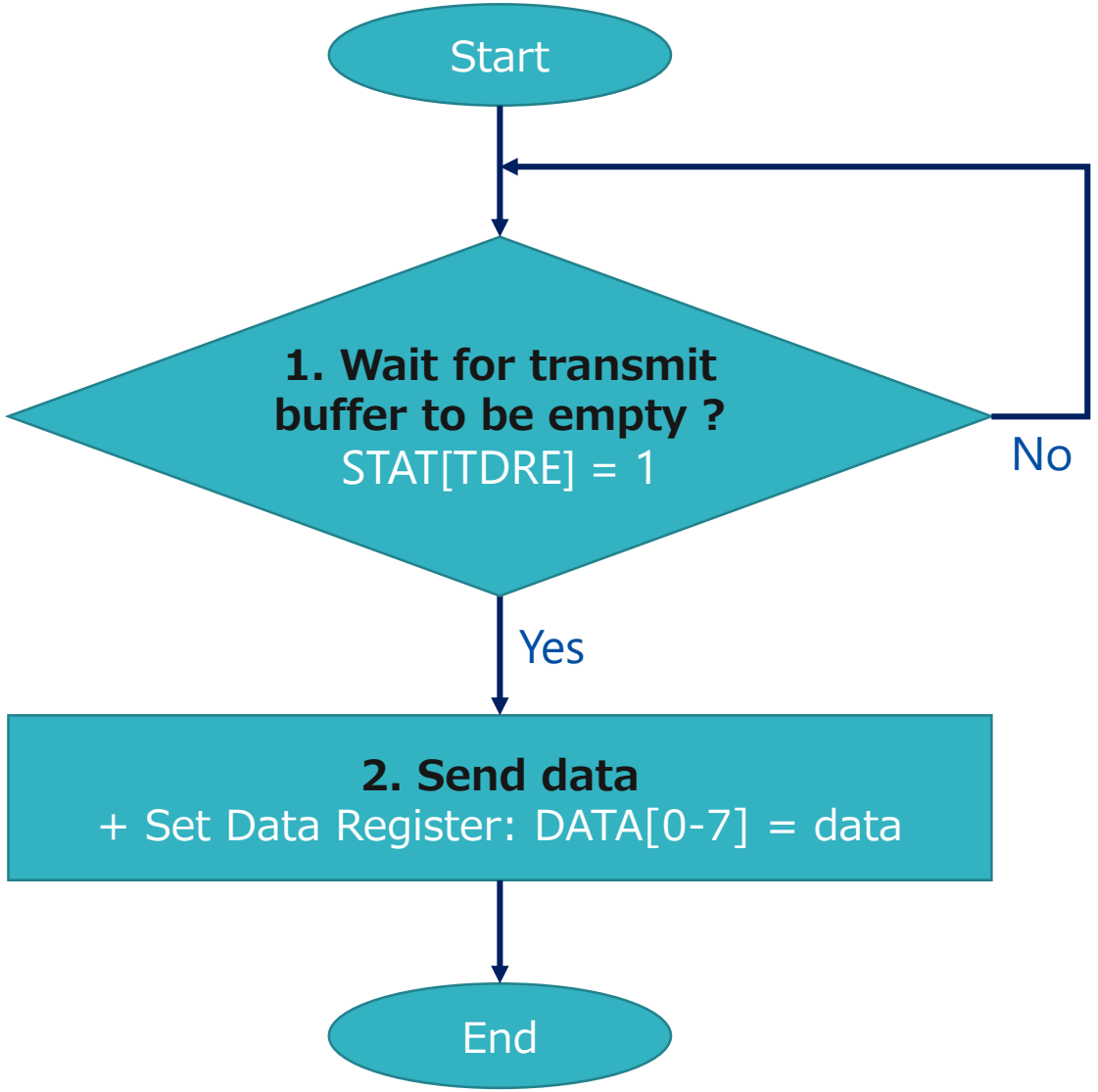
```
typedef struct {
        register
        ……
} LPUART_Type;

#define LPUART0_base_address (0x4006A000u)
#define LPUART0 ((LPUART_Type*)LPUART0_base_address)
```

3.7. Configuration for LPUART



3.8. Configuration for LPUART transmit single Char



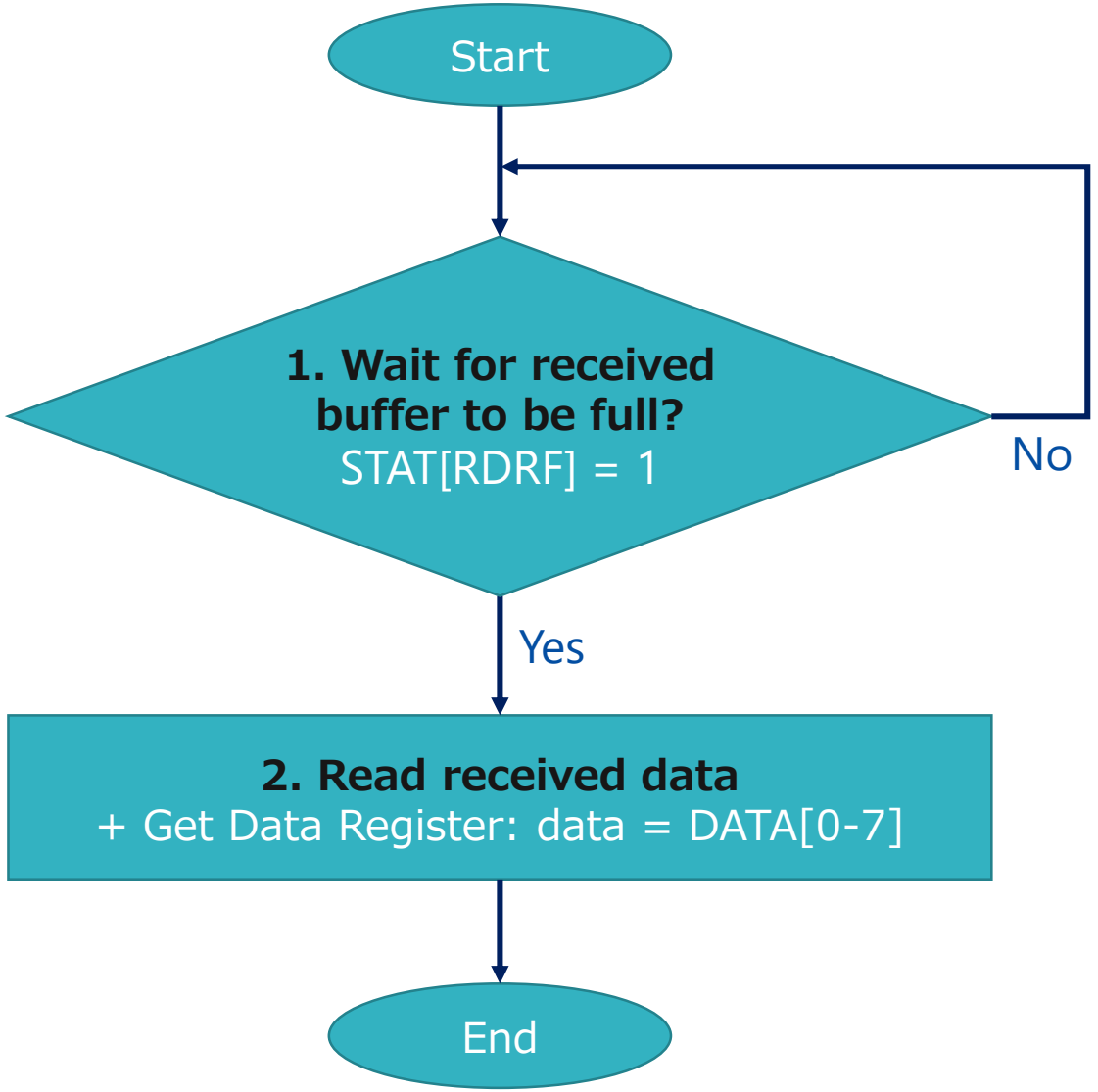
3.8.1. Assignment 2

Write a program to **Transmit** single char via LPUART

- Baud rate: 9600
- 1 stop bit
- 8 bit format
- no parity

- Send char: ‘h’

3.9. Configuration for LPUART Receiver single Char



3.9.1. Assignment 3

Write a program to **Receive** single char via LPUART

- Baud rate: 9600
- 1 stop bit
- 8 bit format
- no parity

- if single char received is ‘1’ then led on
else single char received is ‘0’ then led off

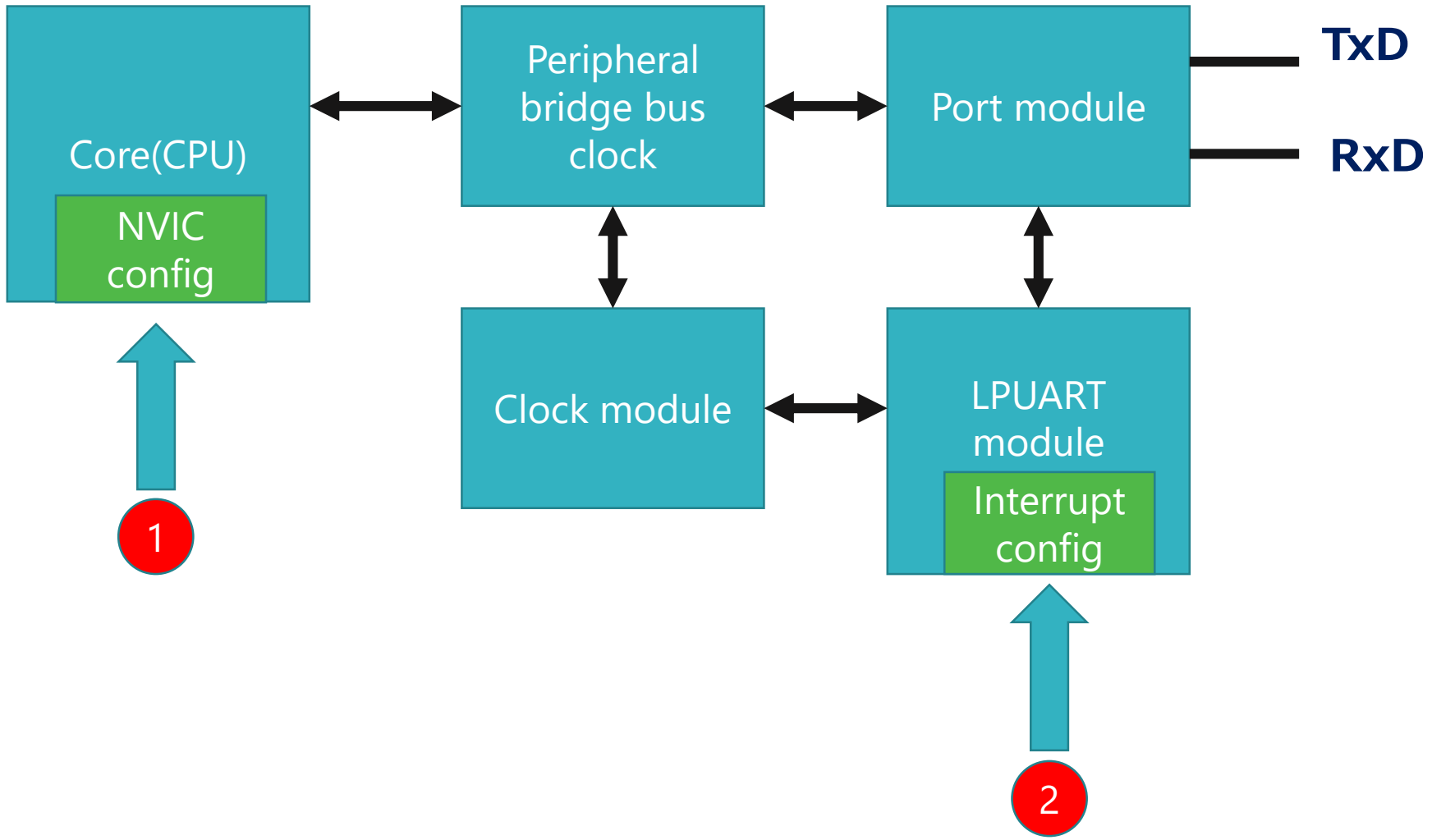
3.10. Assignment 4

Write a program to **Transmit** a string via LPUART

- Baud rate: 9600
- 1 stop bit
- 8 bit format
- no parity

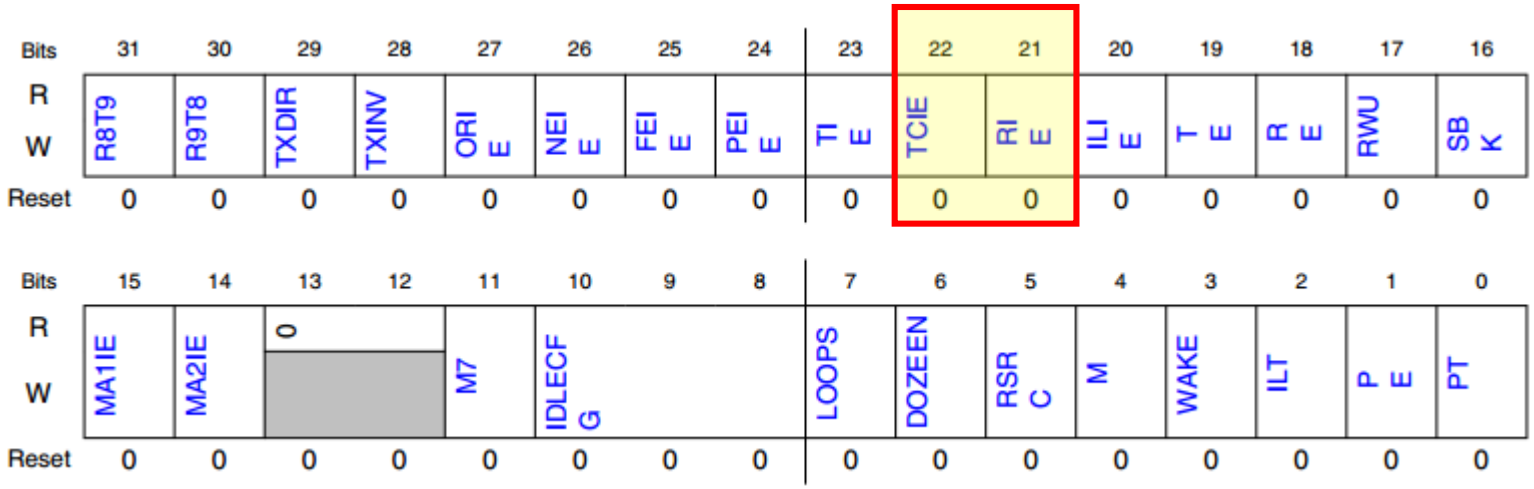
- Send string: “Hello world/r/n”

4. LPUART Interrupt



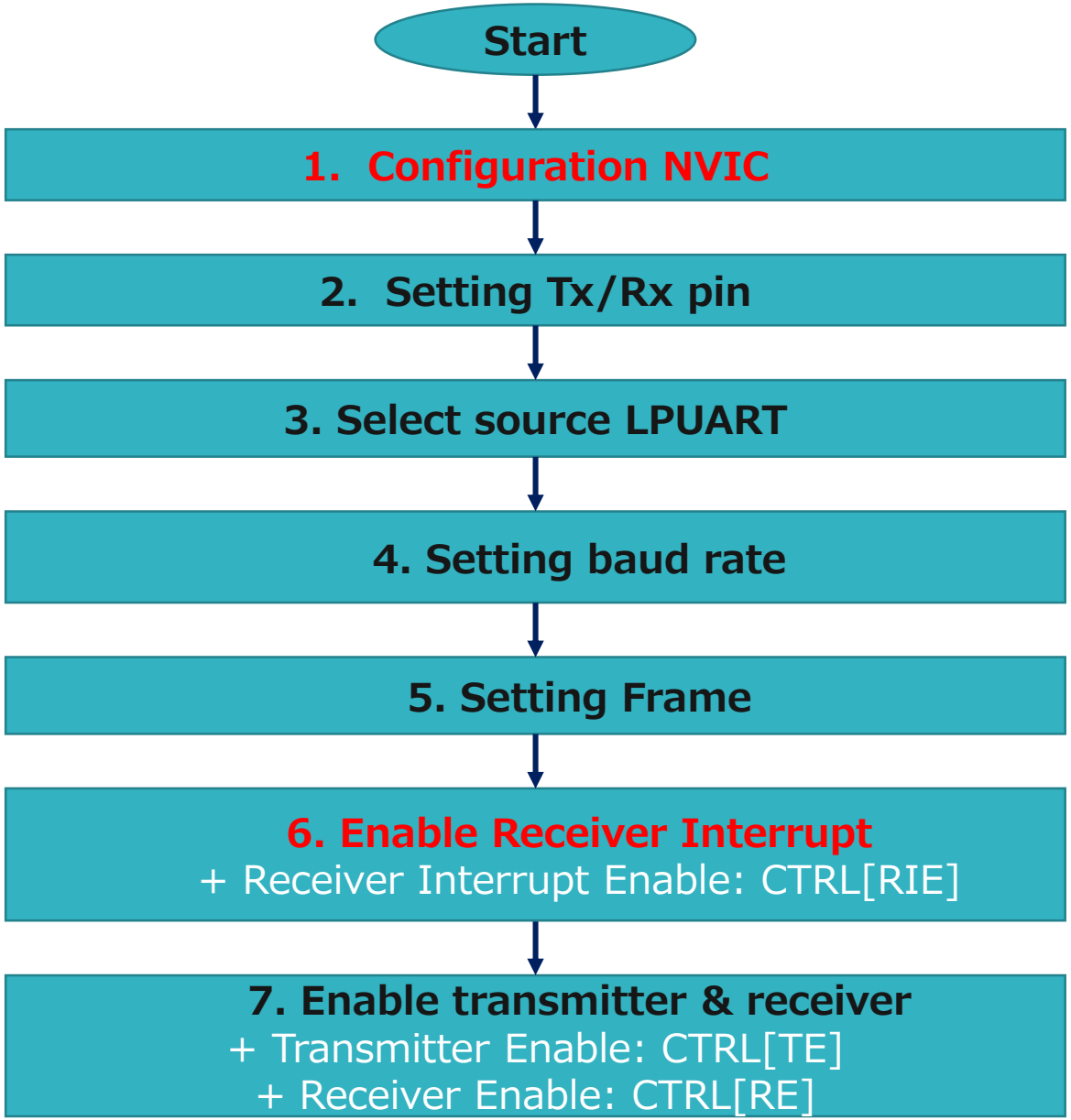
4. LPUART Interrupt

LPUART Control Register (CTRL): Enable Receiver, Transmission Complete Interrupt



Field	Function
22 TCIE	Transmission Complete Interrupt Enable for TCIE enables the transmission complete flag, TC, to generate interrupt requests. 0b - Hardware interrupts from TC disabled; use polling. 1b - Hardware interrupt requested when TC flag is 1.
21 RIE	Receiver Interrupt Enable Enables STAT[RDRF] to generate interrupt requests. 0b - Hardware interrupts from RDRF disabled; use polling. 1b - Hardware interrupt requested when RDRF flag is 1.

4.1. LPUART Receiver Interrupt



4.2. Assignment 5

Write a program **LPUART Receiver interrupt**

- Baud rate: 9600
- 1 stop bit
- 8 bit format
- no parity

- if single char received is '**1**' then led on
else single char received is '**0**' then led off

4.3. Assignment 6

Write a program **LPUART Receiver interrupt**

- Baud rate: 9600
- 1 stop bit
- 8 bit format
- no parity

- if a string received is “[LEDON]” then led on
else a string received is “[LEDOFF]” then led off


Thank you